

BOOKLET I

STRUCTURED QUESTIONS

QUESTION 1

Hepatitis B is an infectious illness caused by the hepatitis B virus. Vaccines, which have been developed for the prevention of hepatitis B viral infections, rely on the use of a viral envelope protein called hepatitis B surface antigen (HBsA). The original vaccine, prepared from plasma obtained from patients who had long-standing hepatitis B viral infections, could be contaminated with the virus. To circumvent the problem, new vaccines were made using recombinant DNA technology. The bacterium *Escherichia coli* is commonly used to produce HBsA for the manufacture of vaccines.

- (a) (i) Apart from preventing infections caused by contaminated vaccines, outline one other advantage of using recombinant DNA technology to produce vaccines against hepatitis B.

[1]

- (ii) State a reason why *E. coli* is commonly used to produce recombinant HBsA used in the vaccines.

[1]

Fig. 1.1 shows part of the hepatitis B viral genome which includes the *HBsA* gene together with the positions of the restriction sites of three restriction enzymes, *EcoRI*, *HindIII* and *HpaI*. These restriction sites do not occur elsewhere within the hepatitis B viral genome.

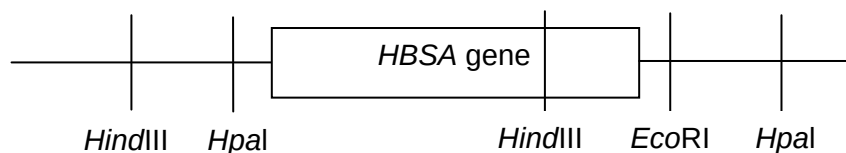


Fig. 1.1

Fig. 1.2 shows the bacterial plasmid pXMAS25 with the positions of the restriction sites of three restriction enzymes, *EcoRI*, *HindIII* and *PstI*, available at its multiple cloning site. These restriction sites do not occur elsewhere within the plasmid pXMAS25.

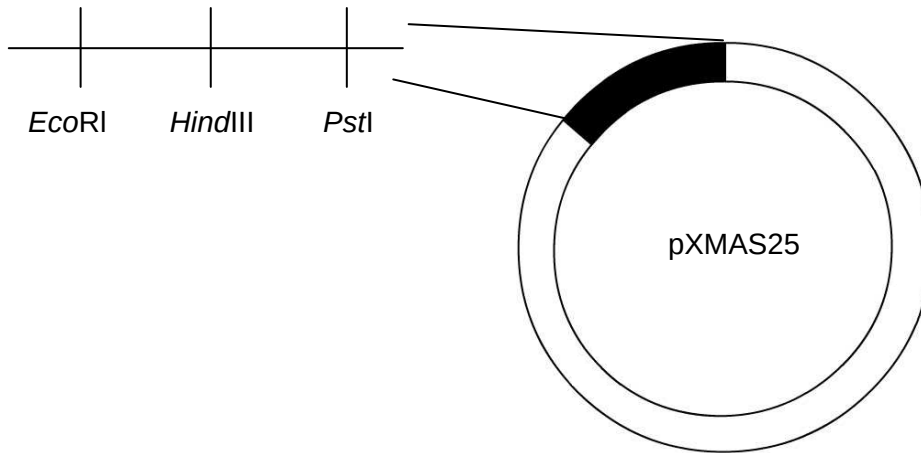


Fig. 1.2

All restriction sites in **Fig. 1.1** and **Fig. 1.2** generate sticky ends when cut, except for the *HpaI* site.

- (b) With reference to **Fig. 1.1** and **Fig. 1.2**, describe the procedure to create a recombinant pXMAS25 plasmid containing the viral *HBSA* gene. [5]

To produce recombinant eukaryotic proteins such as human growth hormone (hGH) using *E. coli* cells, the eukaryotic *hGH* gene cannot be obtained from genomic DNA, unlike the case of the viral *HBSA* gene.

- (c) Explain why this is so, and suggest how the *hGH* gene sequence can be derived from another source to ensure its proper expression in *E. coli*. [3]

In recent years, banana has been successfully engineered to produce HBSA but the banana is often in short supply. A banana tissue culture has been developed for the mass propagation of the banana plant. Different combinations of plant growth regulators are added to banana explants and their effects are shown in **Fig. 1.3**.

Plant growth regulators (mg L ⁻¹)		Percentage of explant differentiated (%)	Mean number of shoots per explant
Auxins	Cytokinins		
0	0	29	2
0.1	1.0	42	4
0.1	2.5	46	8
0.1	5.0	58	8
0.5	1.0	54	12
0.5	2.5	66	20
0.5	5.0	58	28

Fig. 1.3

- (d) With reference to **Fig. 1.3**,
- (i) state whether shoot regeneration from explants requires plant growth regulators. Give an explanation for your answer. [2]

- (ii) identify, with a reason, which combination of growth regulators is recommended for shoot regeneration from banana explants. [2]

Apart from banana, potato is another common plant which can be engineered to produce HBsAg. However, banana is the preferred choice for the production of oral vaccines against hepatitis B.

- (e) Suggest an advantage of using banana over potato to produce oral vaccines against hepatitis B. [1]

[Total: 15 marks]

BOOKLET II

STRUCTURED QUESTIONS (continued)

QUESTION 2

- (a) Outline what is meant by Restriction Fragment Length Polymorphisms (RFLPs). [1]

Linkage analysis was performed on a family with a history of a rare recessive genetic disease to map the position of the disease gene relative to two linked RFLP markers **A** and **B** on chromosome 4.

Affected members of the family are represented as filled symbols in the pedigree in **Fig. 2.1**. DNA isolated from each family member is digested with *Eco*RI and the resulting fragments separated by gel electrophoresis before they are transferred to nylon membranes. The resulting blots are hybridised to two DNA probes which specifically detect three alleles **A1**, **A2** and **A3** of RFLP marker **A** and two alleles **B1** and **B2** of RFLP marker **B**. The band pattern for each individual is also shown in **Fig. 2.1**.

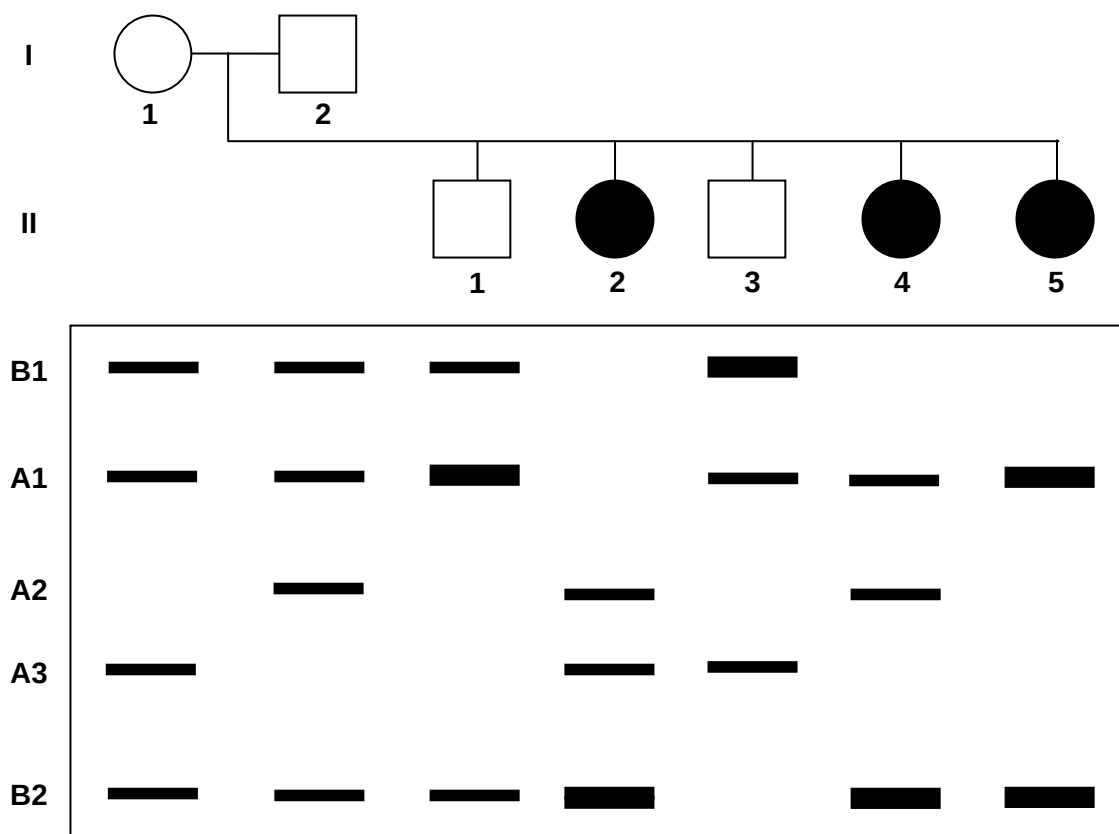


Fig. 2.1

Prior studies on the same family revealed the probable locations of the disease gene and RFLP markers **A** and **B** relative to the centromere on chromosome 4, as shown in **Fig. 2.2**, although it was not yet established which marker is nearer to the disease gene.

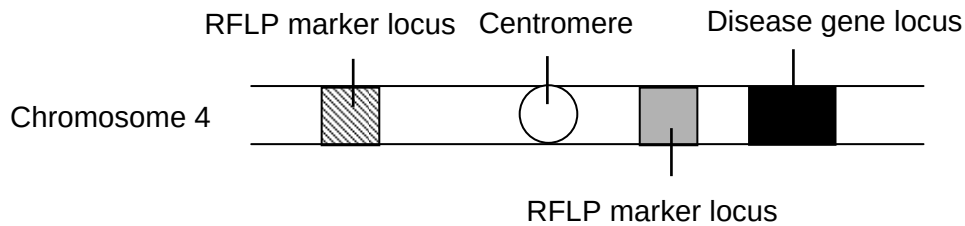


Fig. 2.2

- (b) Based on evidence in **Fig. 2.1**, deduce, with an explanation, whether RFLP marker **A** or **B** is located on the same side of the centromere as the disease gene. [3]

- (c) With reference to **Fig. 2.1** and assuming no mutations occur, suggest why the band pattern of individual **II-2** differs from that of individual **II-4** even though they are both affected with the disease, given that alleles **A1** and **A2** are each linked to the disease allele in individuals **I-1** and **I-2** respectively. [3]

In order to confirm the genotypes of individuals in generation I for the marker **A** locus, RFLP analysis using another restriction enzyme *Hind*III was carried out on individual **I-1**, **I-1**'s brother (not represented in **Fig. 2.1**) and individual **I-2**.

The positions of the *Hind*III site in the three alleles **A1**, **A2** and **A3** are shown in **Fig. 2.3**. The distances between the *Hind*III sites are not shown to scale. The probe used for detection of the RFLPs in Southern blotting binds to the region of the locus as indicated in **Fig. 2.3**.

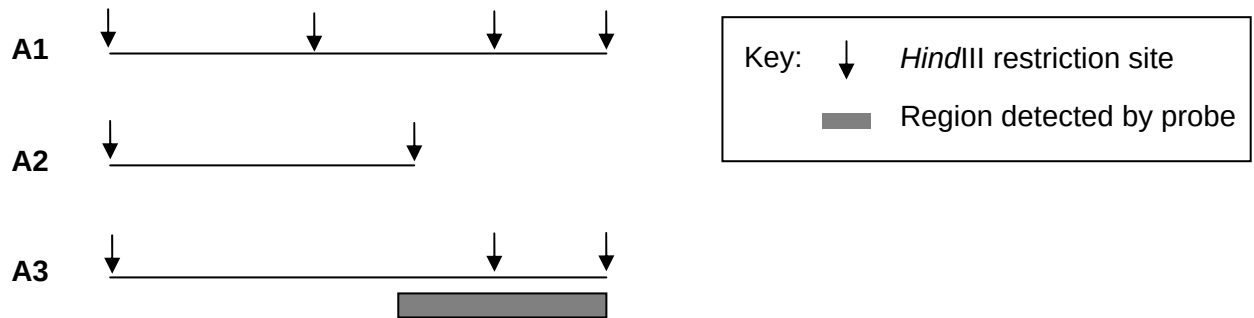


Fig. 2.3

The band patterns obtained for individual **I-1** and her brother are illustrated in **Fig. 2.4**.

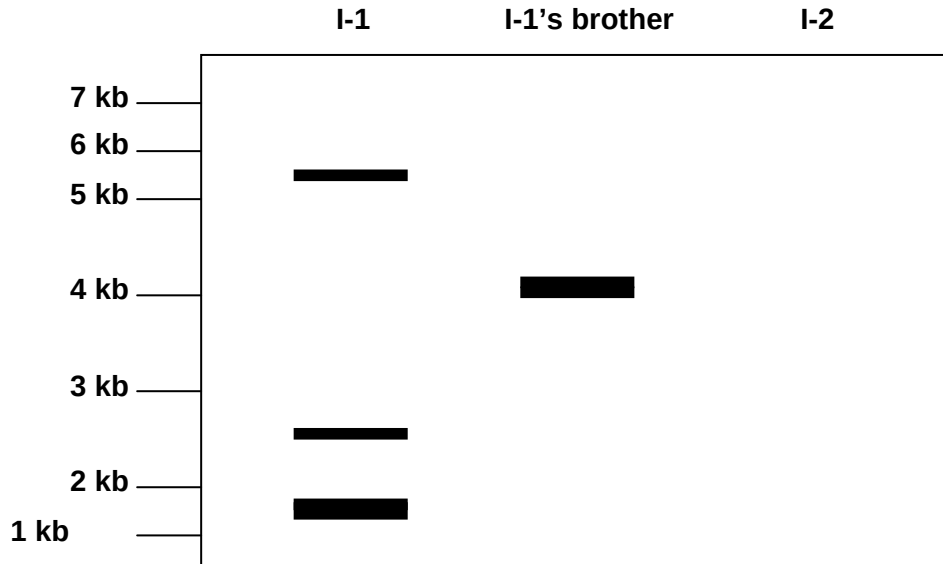


Fig. 2.4

(d) Draw in **Fig. 2.4** the expected band pattern for individual **I-2**.

[2]

Data available from the Human Genome Project (HGP), such as that illustrated in **Fig. 2.5**, is critical for mapping the chromosomal locus of the rare genetic disease by linkage analysis above.

Fig. 2.5 shows a map of other key RFLP markers, besides markers **A** and **B** (which are not shown), in the vicinity of the disease gene locus on chromosome 4. The position of the disease gene locus is also indicated.

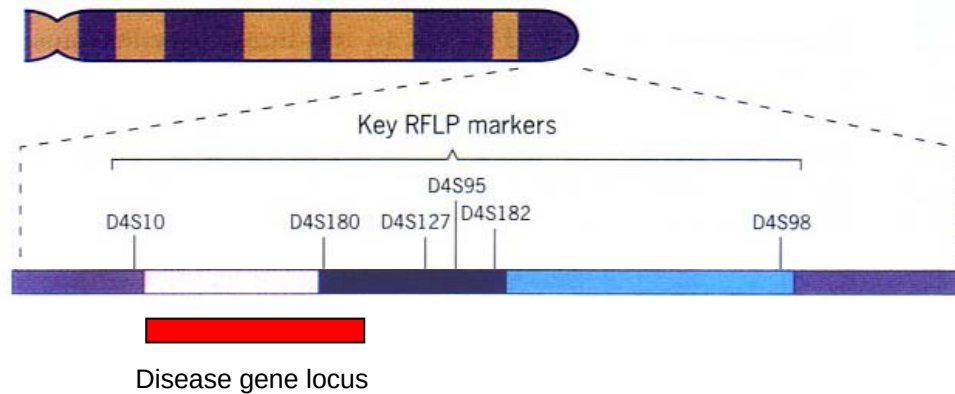


Fig. 2.5

- (e) Determine which other RFLP marker shown in **Fig. 2.5** is to be used to confirm the location of the disease gene locus in this region of chromosome 4. Explain your answer.[2]

- (f) Suggest how information from **Fig. 2.5** can be used to achieve a specified goal of the HGP. [1]

[Total: 12 marks]

NAME: _____

CT GROUP: 10S7 _____

BOOKLET III

STRUCTURED QUESTIONS (continued)

QUESTION 3

Rice, *Oryza sativa*, is usually intolerant of sustained submergence under water, although it grows rapidly in height for a few days before dying. This is true for one variety, *Oryza sativa japonica*. Another variety, *Oryza sativa indica*, is much more tolerant to submergence. Three genetically modified forms of *O. sativa japonica*, GMFA, GMFB and GMFC, were created using different fragments of DNA taken from *O. sativa indica*. The plants were then submerged for a period of 11 days. **Fig. 3.1** shows the measured heights of all the plants at the beginning and end of the submergence period.

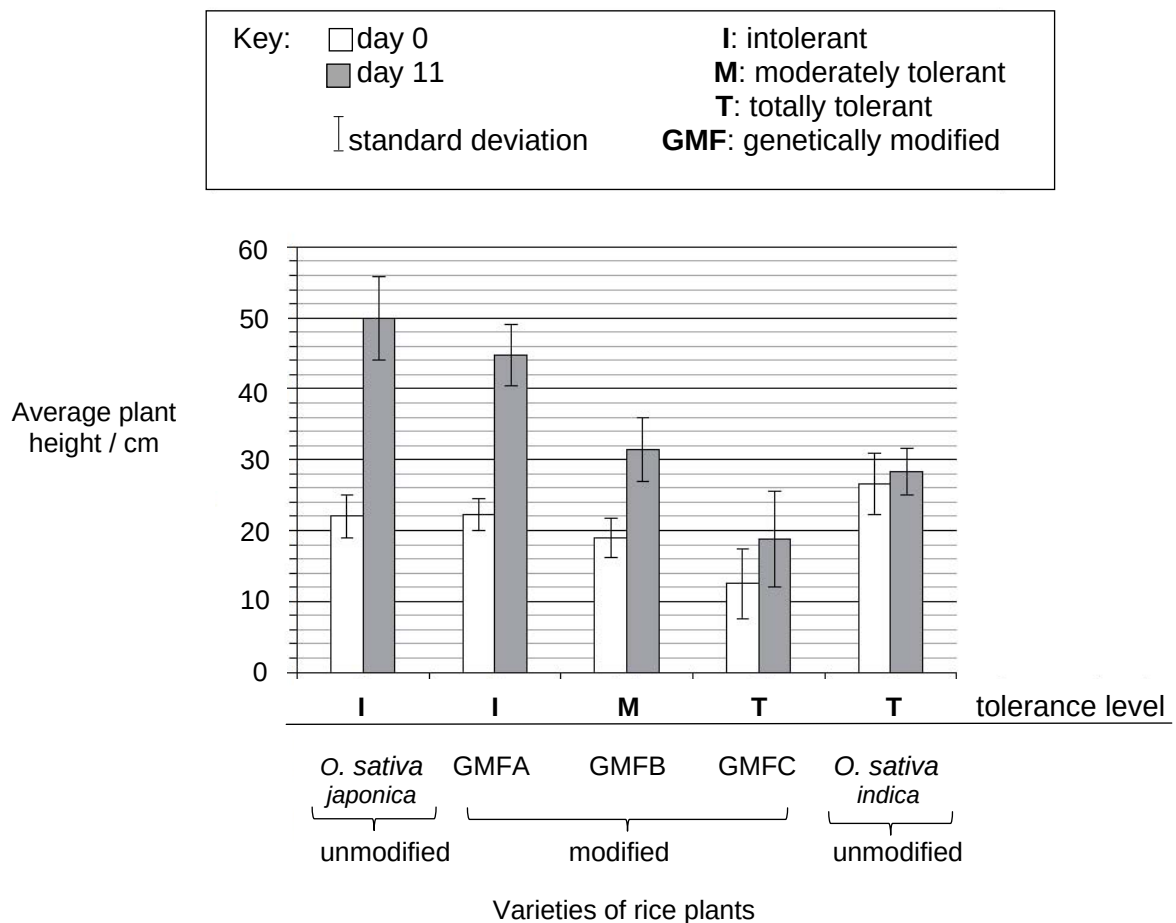


Fig. 3.1

(a) Outline the use of the binomial system of nomenclature in *Oryza sativa*.

[2]

- (b) Describe the general relationship between the growth of all the *japonica* varieties and their stated tolerance levels. [2]

- (c) Suggest how the error bars can be used to compare the results for *O. sativa indica*. [1]

In the same experiment, the researchers hypothesised that the capacity of rice to survive when submerged is related to the presence of three genes, *SubD*, *SubE* and *SubF*. These genes are found very close to each other on rice chromosome number 9. **Fig. 3.2** shows part of a gel indicating the relative amounts of mRNA produced from these three genes by the submergence-intolerant variety *O. sativa japonica* and by the submergence-tolerant variety *O. sativa indica* at different times of a submergence period, followed by a recovery period out of water.

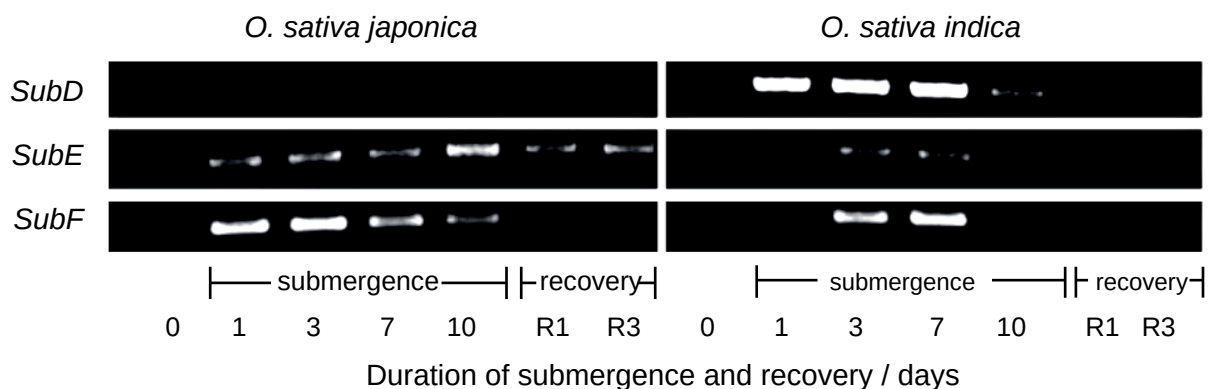


Fig. 3.2

- (d) Compare mRNA production from the three genes between the two varieties during the submergence period. [3]

(e) With reference to **Fig. 3.1** and **Fig. 3.2**,

(i) deduce which gene was used to modify GMFC. Give a reason for your answer. [2]

(ii) suggest a rationale for including the recovery period in this experiment. [1]

(iii) evaluate how modified varieties of rice could be used to overcome food shortage problems in some countries. [2]

[Total: 13 marks]

PLANNING QUESTION

QUESTION 4

Freshwater crabs are land-dwelling organisms. Owing to their relatively poor dispersal abilities, they are easily isolated by geographical barriers such as seas and rivers. Recently, specimens of *Sayamia* were collected from two rivers in Taiwan. *Sayamia* species are native to southern Indochina (Thailand, Cambodia and southern Vietnam) and northern Peninsular Malaysia, and have never been reported from Taiwan.

Plan an investigation to construct a phylogeny for these newly found *Sayamia* species and their native populations in southern Indochina and northern Peninsular Malaysia. Comparison should be made using the 16S rRNA gene (1.5 kb) and cytochrome oxidase gene (0.7 kb).

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- tissue samples from *Sayamia* collected from Taiwan and native populations in southern Indochina and northern Peninsular Malaysia.
- tissue samples from *Somanniathelphusa taiwanensis* and *Siamthelphusa improvisa* to be used as outgroups
- pestle and mortar
- DNA extraction buffer solution
- PCR machine and *Taq* polymerase
- DNA primers flanking the 16S rRNA and cytochrome oxidase genes
- microcentrifuge tubes
- centrifuge
- agarose gel
- ethidium bromide
- automated DNA sequencing machine (DNA sample can be loaded and its sequence determined)
- computer with analysis software

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure
- a description of the method used including scientific reasoning behind the method
- the type of data generated by the experiment
- how the results will be analysed including how phylogeny of these newly found *Sayamia* species and their native population in southern Indochina and northern Peninsular Malaysia can be determined.

[Total: 12 marks]

FREE RESPONSE QUESTION

Answer the question.

Your answers must be in continuous prose, where appropriate.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required.

QUESTION 5

- (a) Compare between embryonic stem cells and blood stem cells. [6]
- (b) Explain two approaches by which stem cells can be used to treat Severe Combined Immunodeficiency Disease (SCID). [8]
- (c) Discuss the social and ethical considerations in the use of gene therapy. [6]

[Total: 20 marks]

--- END OF PAPER---